

Exploratory Data Analysis Summary

Aspect Based Sentiment Analysis for Laptop Reviews

1. Dataset Overview

This analysis was performed on the cleaned dataset produced in the previous stage, containing 16,991 reviews across 365 unique laptop products and 20 distinct brands, following removal of duplicate rows and correction of numeric data types.

2. Scope of Analysis

The analysis covered univariate analysis of rating and review length, bivariate analysis of rating against review length, brand, and product popularity, multivariate analysis combining brand and popularity tier together, and correlation analysis across all numeric fields. Time-series analysis was not performed, since the dataset does not contain any date or timestamp column, this is noted as a limitation of the available data.

3. Key Business Insights

- The overall rating distribution is heavily skewed toward positive reviews, 58 percent of reviews are 5 star, while only about 10 percent are 1 or 2 star combined.
- About 3.7 percent of reviews are likely truncated by the web scraper, ending mid-sentence with a literal READ MORE marker, a limitation relevant to future text based analysis.
- Review length shows almost no linear relationship with star rating overall, with a correlation of -0.04. The main observable pattern is that 5 star reviews tend to be somewhat shorter on average than 1 to 4 star reviews.
- Primebook, Apple, and realme have the highest average ratings among brands with sufficient review volume, while walker, Ultimus, and ZEBRONICS have the lowest.
- The four most reviewed brands, HP, ASUS, Acer, and Lenovo, all sit in the middle of the overall rating range, performing decently but not exceptionally compared to smaller brands.
- Products with very few ratings show highly variable overall ratings, ranging from 3.3 to 5.0, while products with thousands of ratings settle into a stable range of 4.2 to 4.4, indicating that rating reliability increases with review volume.
- Among all numeric variables, only no_ratings and no_reviews show a strong correlation with each other, at 0.91, since both reflect overall product popularity rather than two independent factors. All other numeric relationships in the dataset are weak.
- For most well established brands, high popularity products are rated equal to or higher than low popularity products, though several brands show an unexplained dip specifically in the medium popularity tier, worth further investigation at the individual product level.

4. Recommendations

- When building and evaluating the sentiment extraction model in later stages, account for the imbalance toward positive reviews, particularly by checking performance on negative reviews specifically, not just overall accuracy.
- Treat product ratings based on very few reviews with caution when comparing products, since these are statistically less reliable than ratings based on large review volumes.
- Investigate the specific complaints behind lower rated brands such as walker and Ultimus in the upcoming aspect extraction stage, to identify concrete, actionable product issues.
- Since numeric variables show weak correlations overall, prioritize the review text itself as the primary source of insight going forward, supporting the core direction of the project.
- Further investigate the medium popularity tier dip seen in brands like DELL and Acer at the individual product level, rather than treating it as a general brand wide pattern.

5. Limitations

This dataset does not include a date or timestamp field, so no time based trend or seasonality analysis could be performed. A small portion of reviews, approximately 3.7 percent, appear truncated due to scraper limitations. Brand level comparisons excluded brands with fewer than 30 reviews, since averages from very small samples were considered unreliable.

6. Conclusion

The exploratory analysis confirms that numeric fields alone, such as rating counts and review counts, provide only weak signals about product quality or customer sentiment. This supports the core objective of the project, extracting sentiment directly from review text at the aspect level, which is expected to surface far richer and more actionable insight than the summary statistics examined in this stage.